Electron-Ion Path Integral Monte Carlo with the Pair-Product Action

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I. INTRODUCTION

In order to study the quantum statistical properties of the H fluid mixture through a computer experiment it is necessary to add to our simulation box of volume Ω an equal number $N/2$ of electrons (e) of mass m_e and of protons (p) of mass m_p and let them interact with the usual Coulomb pair potential $\pm 1/r$. We thus obtain a fluid of number density $n = N/\Omega$. We can then study the properties of H in thermal equilibrium at an inverse temperature $\beta = 1/k_B T$ with k_B Boltzmann constant and T the absolute temperature. If the laws of quantum statistical physics and of Coulomb are meaningful we should be able to reproduce the thermodynamic and structural properties of H that we observe in nature.

In our reduced units we will measure lengths in units of a Bohr radius $a_B = \hbar^2/m_e e^2$ and energies in units of two Rydbergs e^2/a_B, i.e. of a Hartree. The Hamiltonian operator and the density matrix in reduced units will then read

$$H = K + V = -\sum_{\alpha=1}^{N} \lambda_{i_\alpha} \nabla_\alpha^2 + \sum_{\alpha<\beta=1}^{N} \phi_{i_\alpha i_\beta}(r_{\alpha\beta}), \tag{1.1}$$

$$\rho = e^{-H/T}. \tag{1.2}$$

where we indicate with a Greek index the particle label and with a Roman index the species label so that for example i_α is the species label of particle α, either 1 for the electrons or 2 for the protons, $\boldsymbol{r}_\alpha$ is the position vector of particle α, and $r_{\alpha\beta} = |\boldsymbol{r}_{\alpha\beta}| = |\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta|$ is the distance between particles α and β. The partial pair potential is the usual Coulomb potential $\phi_{ij} = \epsilon_{ij}/r$ with $\epsilon_{ii} = +1$ for all i and $\epsilon_{ij} = -1$ for $i \neq j$. $\lambda_i = \hbar^2/2\mu_i$ where the reduced masses are $\mu_1 = 1$ and $\mu_2 = m_p/m_e \approx 1836.15$. Temperatures are in units of ≈ 315775K.

The coordinate representation of the high temperature density matrix for distinguishable particles is [1]

$$\rho(R, R'; \tau) = \rho_0(R, R'; \tau) e^{-U(R,R';\tau)}, \tag{1.3}$$

$$\rho(R, R'; \tau) = \prod_{\alpha=1}^{N} (4\pi\lambda_{i_\alpha}\tau)^{-3/2} e^{-(\boldsymbol{r}_\alpha - \boldsymbol{r}'_\alpha)^2/4\lambda_{i_\alpha}\tau}, \tag{1.4}$$

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where $\tau = \beta/M$ for large M. So that the coordinate representation of the thermal density matrix for identical particles can be obtained from the following discretized path integral [2]

$$\rho(R, R'; \beta) = \sum_{\mathscr{P}} \mathrm{sgn}(\mathscr{P}) \int \rho(\mathscr{P}R, R_1; \tau)\rho(R_1, R_2; \tau) \cdots \rho(R_{M-1}, R'; \tau) \prod_{k=1}^{M-1} dR_k, \tag{1.5}$$

where $R_k = (\boldsymbol{r}_{1,k}, \boldsymbol{r}_{2,k}, \ldots, \boldsymbol{r}_{N,k})$ is the kth timeslice. Here $\boldsymbol{r}_{\alpha,k}$ is the position vector of particle α of species i_α at timeslice k in the path integral. The sum is over the permutations $\mathscr{P}$ of identical particles (of the same species) and $\mathrm{sgn}(\mathscr{P})$ is the sign of the given permutation

$$\mathrm{sgn}(\mathscr{P}) = \begin{cases} +1 & \text{Bose-Einstein statistics} \\ \begin{cases} +1 & \mathscr{P} \text{ an even permutation} \\ -1 & \mathscr{P} \text{ an odd permutation} \end{cases} & \text{Fermi-Dirac statistics} \end{cases} \tag{1.6}$$

While a Monte Carlo integration [3] of the path integral (1.5) (PIMC) can be carried out exactly for bosons, for fermions the alternating signs in the permutation sum becomes less and less efficient as N and/or M increase [4]. This is known as the 'fermions sign problem'. An approximate way to circumvent this problem is the Restricted Path Integral Monte Carlo (RPIMC) method of Ceperley [2, 4]. In our simulations we will always compare the PIMC for bosons and the RPIMC for fermions with a free fermions restriction.

For the inter-action U we will adopt the end-point approximation for the pair-product action [1]

$$U(R, R'; \tau) \approx \sum_{\alpha<\beta} P_{i_\alpha i_\beta}(\boldsymbol{r}_{\alpha\beta}, \boldsymbol{r}_{\alpha\beta}; \tau), \tag{1.7}$$

where for P_{ij} we choose the pair Coulomb density matrix of Pollock [5]. Namely

$$P_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau) = -\ln \left[\frac{\rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}{\rho^0_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)} \right], \tag{1.8}$$

where

$$\frac{\partial \rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau)}{\partial \tau} = \left[\lambda_{ij} \boldsymbol{\nabla} - \frac{\epsilon_{ij}}{r} \right] \rho_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau). \tag{1.9}$$

Here $\lambda_{ij} = \lambda_i + \lambda_j$ and $\rho^0_{ij}(\boldsymbol{r}, \boldsymbol{r}'; \tau) = (4\pi\lambda_{ij}\tau)^{-3/2} \exp[-(\boldsymbol{r} - \boldsymbol{r}')^2/4\lambda_{ij}\tau]$ satisfies the same Eq. (1.9) but for $\epsilon_{ij} = 0$ for all i, j.

TABLE I. Cases treated in our simulations for the H fluid mixture. We show the total number of particles N, the reduced number density $n = N/\Omega$, the reduced temperature T, the total kinetic energy per particle $e_K = \langle K \rangle/N$, the total potential energy per particle $e_V = \langle V \rangle/N$, the superfluid fraction for species i, $f_s(i)$. In the statistics column "b" stands for bosons, "f" for fermions. The simulation lasted more than 10^8 Monte Carlo steps.

case	statistics	T	M	n	N	Ne_K	$-Ne_V$	$f_s(1)$	$f_s(2)$
A	b (PIMC)	0.15	20	0.3	30	...	...	...	...
B	f (RPIMC)	0.15	20	0.3	20	...	...	...	...
A	b (PIMC)	0.15	20	0.16	30	...	...	...	...
B	f (RPIMC)	0.15	20	0.16	20	...	...	...	...

Appendix A: Description of our PIMC and RPIMC algorithms

In the RPIMC we reject the move whenever the product $D^0_1 D^0_2$ changes sign, where

$$D^0_i = \det \left|\left| (4\pi\lambda_i\tau)^{-3/2} e^{-(\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta)^2/4\lambda_i\tau} \right|\right|_{\alpha\beta}, \tag{A1}$$

is the order $N/2$ determinant such that $i_\alpha = i_\beta = i$.

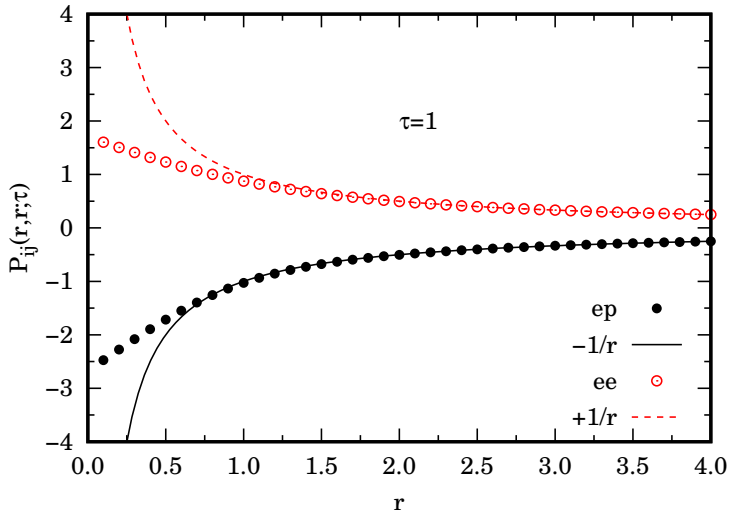

FIG. 1. We show the *ee* and *ep* pair Coulomb density matrix of Eq. (1.8) for $\tau = 1$. We compare it with the bare Coulomb potential. From the figure we can see that the cusp condition [5] $P_{ij}(\boldsymbol{r}, \boldsymbol{r}; \tau) \sim r\epsilon_{ij}/2\lambda_{ij}$ is satisfied at small r. The distance r is in units of a Bohr radius.

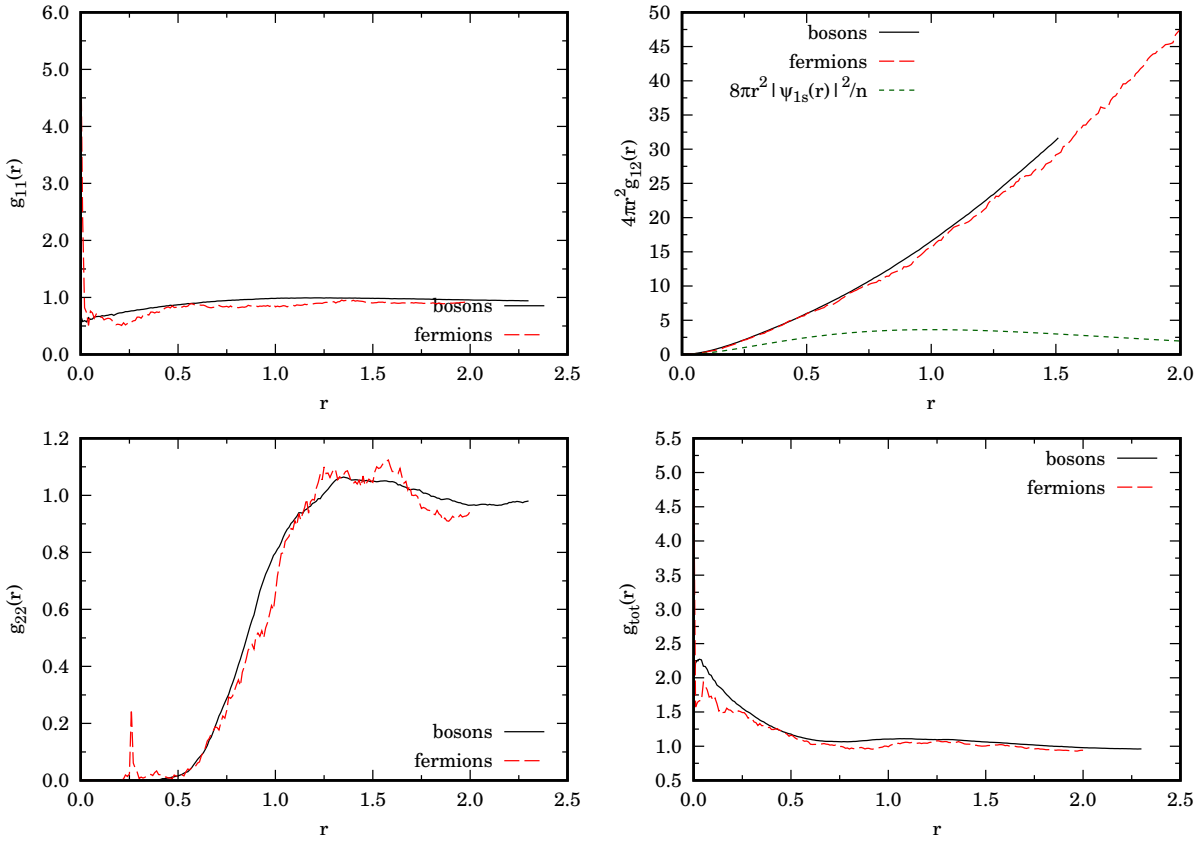

FIG. 2. We show the partial and total RDF of the H fluid for cases A and B in Table I. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

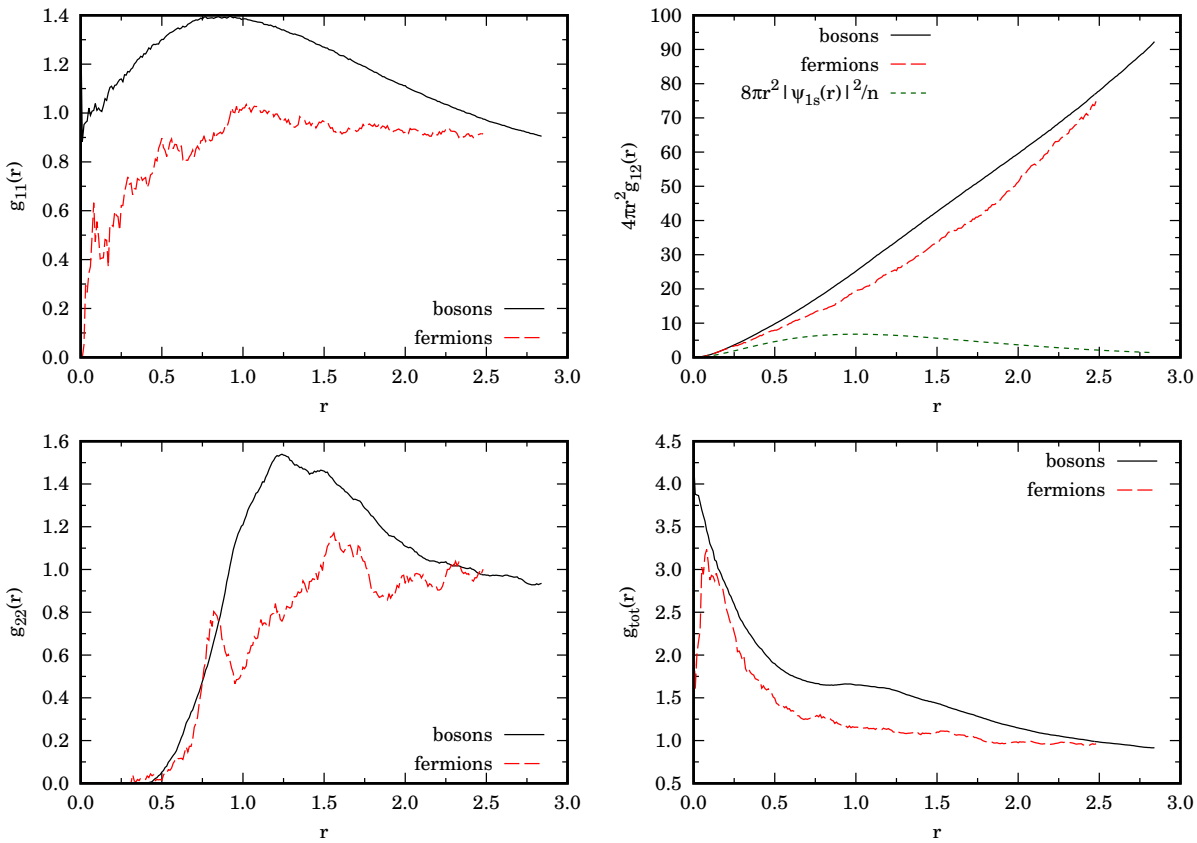

FIG. 3. We show the partial and total RDF of the H fluid for cases C and D in Table I. In the panel for the unlike radial distribution function we also show the ground state probability distribution of the H atom $|\psi_{1s}|^2$ for comparison. The distance r is in units of a Bohr radius.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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None declared.

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